Class-11th
Chapter- Relation & Function
Hand Written Notes

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Class-11th

Cartesian Product :-

The cartesian product of two sets A, B is non-empty set of all ordered pair (a, b) where : $a \in A$ and $b \in B$. This denotes by $A \times B$.

$$A \times B = \{(a,b) : a \in A \text{ \& } b \in B\}$$

$$B \times A = \{(b,a) : a \in A \text{ \& } b \in B\}$$

Ex → $A = \{1, 2\}$, $B = \{2, 3\}$

$$A \times B = \{(1,2), (1,3), (2,2), (2,3)\}$$

Ordered & Unordered Pair :-

Ordered pair = Pair & order → order is important

Unordered pair = pair → order is not important

Ex → Unordered pair same $\{(2,3), (3,2)\}$ different w.r.t ordered pair

Important Points :-

1) $A \times B \neq B \times A$ (in general)
2) If $A \times B = B \times A$ then $A = B$
3) If $n(A) = p$ & $n(B) = q$ then $n(A \times B) = p \cdot q$
4) $A \times \phi = \phi$
5) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
   $A \times (B \cap C) = (A \times B) \cap (A \times C)$
6) $R^2 = R \times R = \{(x,y) : x, y \in R\}$ in plane
7) $R^3 = R \times R \times R = \{(x,y,z) : x, y, z \in R\}$ in 3-D
8) $A \times B \times C = \{(a,b,c) : a \in A, b \in B, c \in C\}$

Relation :-

Relation defined from set A to set B ($R: A \to B$) is subset of Set $A \times B$.

→ Relation defined from $R: A \to B$ is a subset of $(A \times B)$

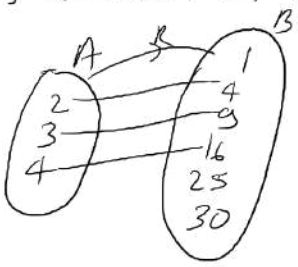
→ Relation defined from $R: B \to A$ is a subset of $(B \times A)$

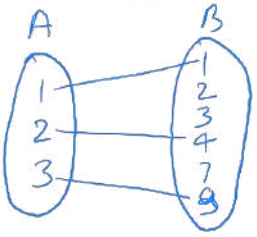
→ Relation defined "on set A" $(R: A \to A)$ is a subset of $(A \times A)$

Ex → A is $\{2, 3, 4\}$, B is $\{1, 4, 9, 16, 25, 30\}$

if $a R b \to b$ is square of a.

$(R: A \to B) = \{(2,4)(3,9)(4,16)\}$



Roaster form	Set Builder form	Arrow diagram
$R = \{(1,1)(2,4)(3,9)\}$ ←	Ex → $A = \{1,2,3\}$, $B = \{1,2,3,4,7,9\}$, $R = \{(a,b) : a^2 = b, a \in A, b \in B\}$	→ 

Domain, Range & co-domain of Relation :-

If $R: A \to B$ then :

i) **Domain :-** Set of element of set A which are related to elements of set B.

ii) **Range :-** Set of element of set B which are related to element of set A.

iii) **Co-domain :-** Complete Set B.

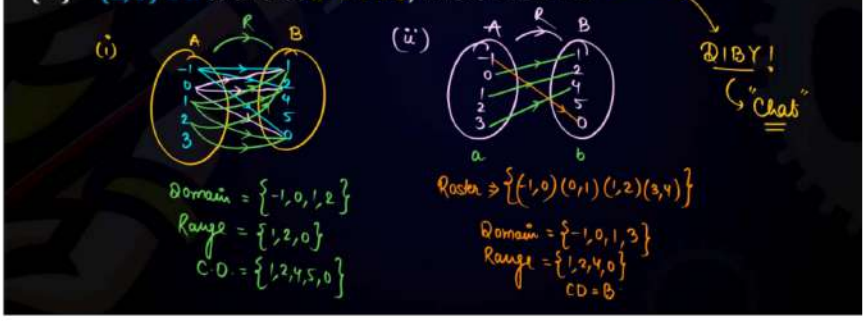
Question

Consider $A = \{-1, 0, 1, 2, 3\}$, $B = \{1, 2, 4, 5, 0\}$, then find domain & range of following relations :

~~(i)~~ $R = \{(a, b) : a^2 + b^2 \le 5, a \in A, b \in B\}$

(ii) $aRb \Leftrightarrow b - a = 1$, where $R : A \to B$

(iii) $(a, b) \in R$ such that $a^2 + 1 = b$, where $R \subset A \times B$.



Important Points :-

1) If $n(A) = p$ and $n(B) = q$ then:

a) number of relations defined from set A to set B $= 2^{pq}$

b) number of relations defined from set B to set A $= 2^{pq}$

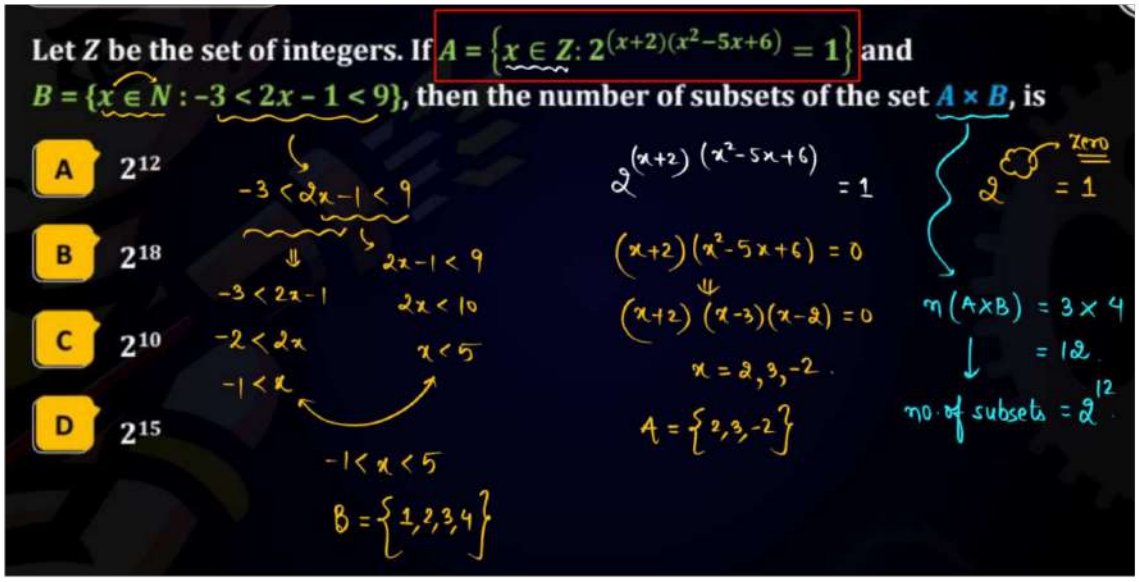
c) Number of relations defined on set A $= 2^{p^2}$

d) Number of relations defined on set B $= 2^{q^2}$

e) Number of non-empty relations defined from set A to set B $= 2^{pq} - 1$

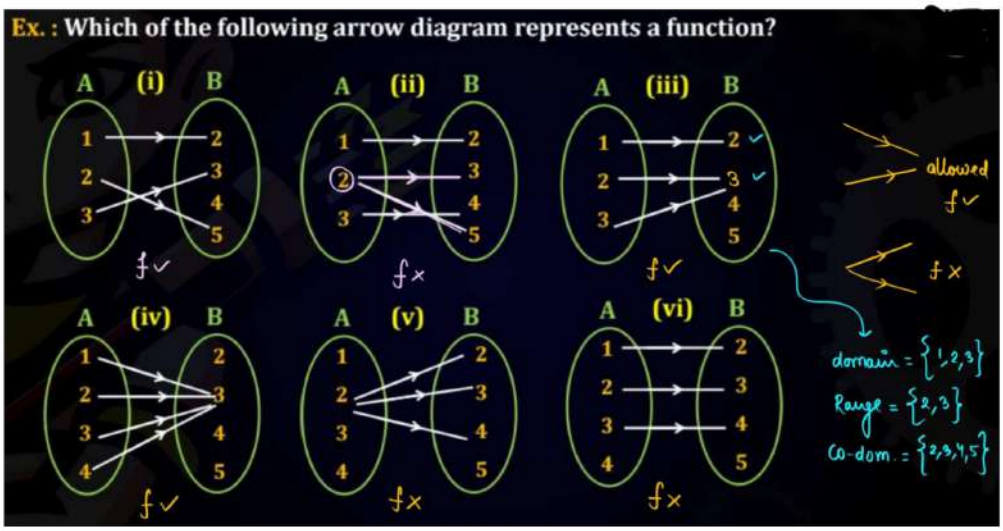
2) Empty relation ↳ $\{\}$

3) Universal Relation :- Square element of A related to square element of set B.



Functions :-

Functions (mapping) is a relation defined from set A to set B ($f : A \to B$) such that all elements of set A to related to only single (one/unique) element in set B.



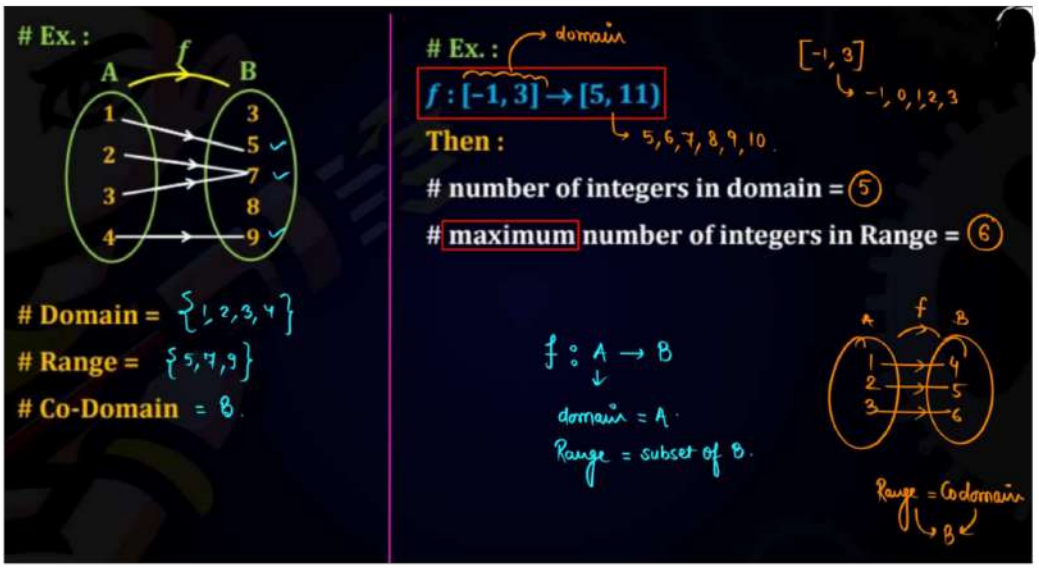
Domain, Range & co-domain :-

If function 'f' is defined from set A to set B ($f: A \to B$) then:

I) Domain of f :- Complete set A.

II) Range of f = element of set B related to elements of set A.

III) Co-domain of f = Complete Set B.



Note :-

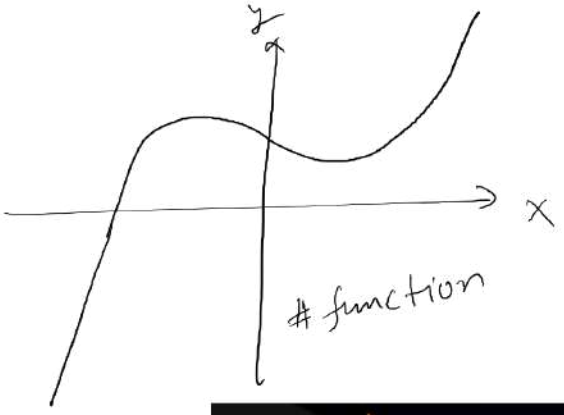
Domain = Set of pre-image / values of x / input / X-axis covered by graph

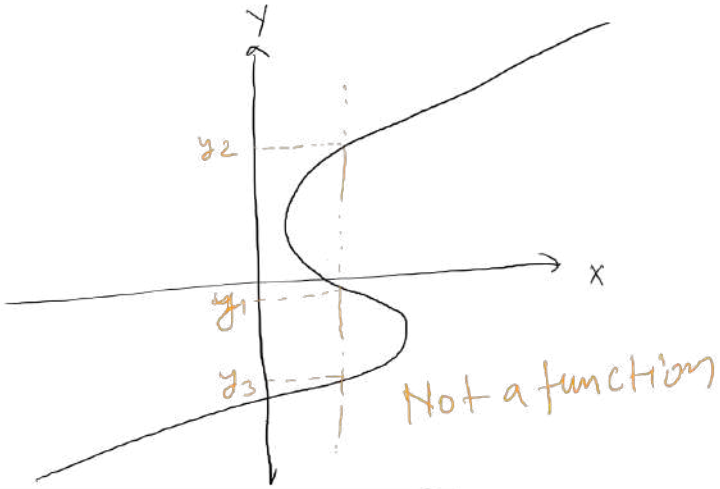
Range = set of functional image / values of y / output / Y-axis covered by graph.

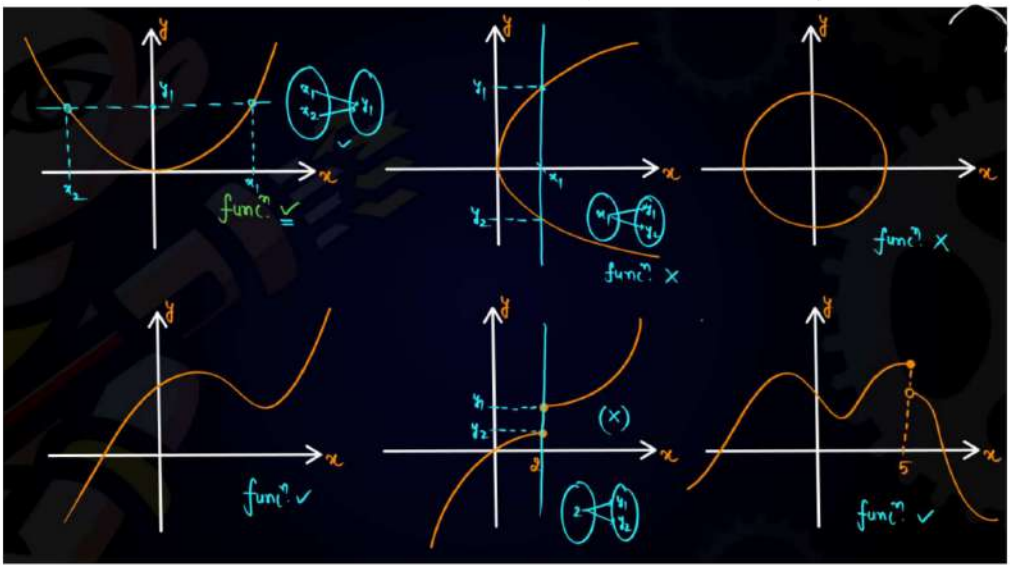
Vertical Line Test :-

In given graph

Draw all possible lines parallel to Y-axis.
If any line cuts the graph at two or more than two points then it is Not a function.





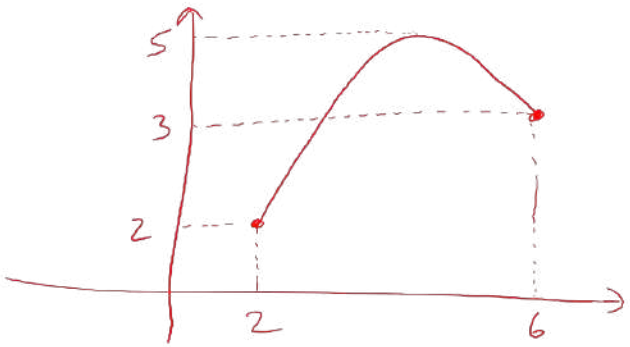


Domain & Range from Graph :-

for any given graph of functions :-

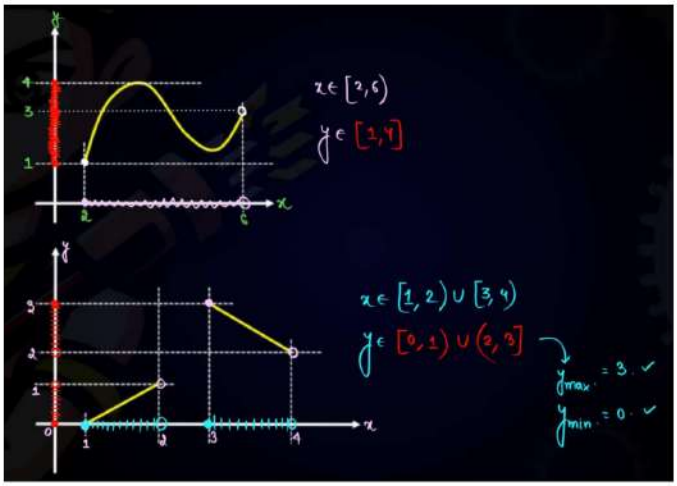
Domain :- portion of x-axis covered by graph.
Range :- portion of y-axis covered by graph.

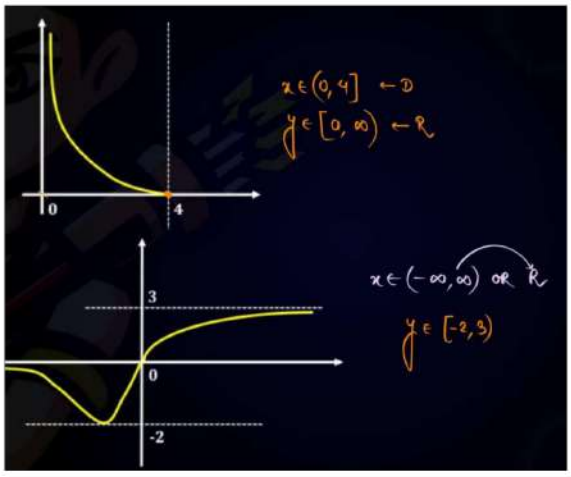
} "only graph of functions"



Domain = $x \in [2, 6]$

Range = $y \in [2, 5]$

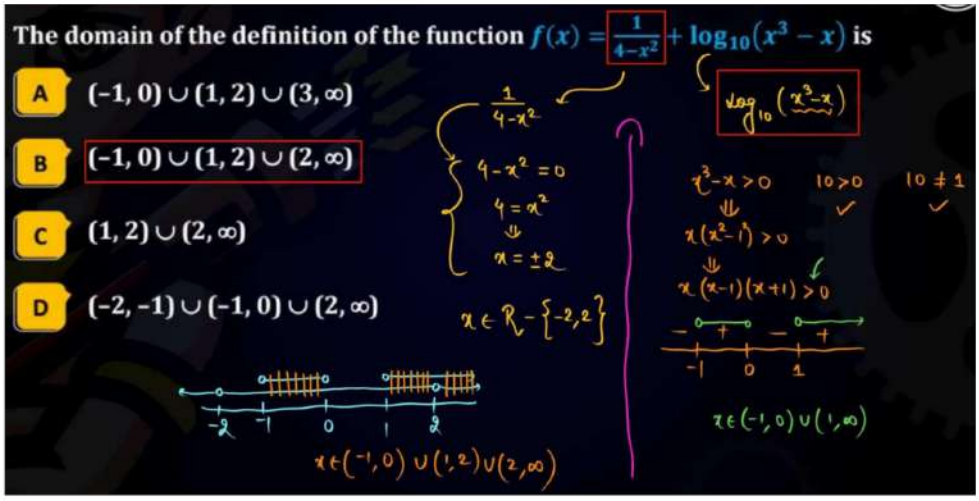




Domain :-

Rules to find domain :-

1) Denominator ≠ 0
   (Find values of x for which denominator = 0 and remove them from Real no set.)
2) Expression under even square root should be greater than or equal to zero.
3) Take care of personal domain of functions
   like - log, Trigonometric functions etc.



Classification of functions :-

1) Polynomial Functions :-

$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, all powers of x = whole no

Ex-

$f(x) = 3x^3 + 5x^2 - 2x + 1$ → degree = 3 cubic poly

Leading coefficient = 3

Note:-

1) Graph of polynomial is always continuous
2) Range of an odd degree polynomial is complete real number set-R
3) Polynomial with leading coefficient '1' are called Monic Polynomial.

Important Property :-

If a polynomial function satisfying

$$f(x)\cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right), \quad x \neq 0$$

then

$$f(x) = 1 \pm x^n, \quad n \in N$$

ii) Rational Function :-

The ratio of two polynomial is called rational function.

Ex → $y = \dfrac{x^{12} + x^2 - 1}{x^6 + x^2 - 1}$, $f(x) = \dfrac{2x-3}{x^2+1}$

iii) Irrational Function! -

Function with operation of addition, subtraction, mutliplication, division and raising to power with non-integral exponent are called irrational functions.

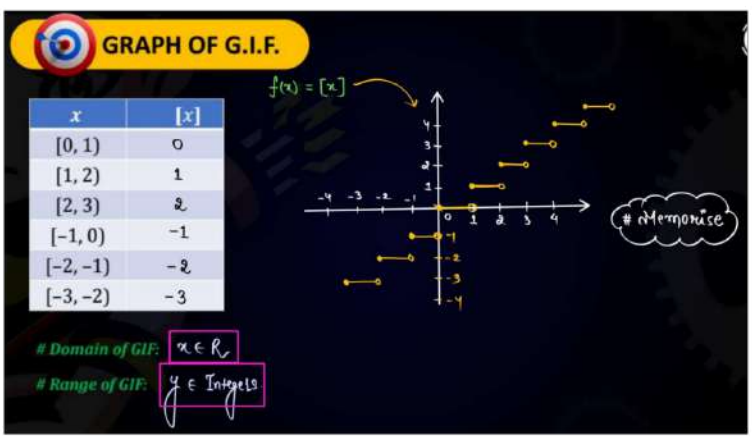
Ex → $y = \sqrt{x^2+5x}$, $y = \dfrac{\sqrt{x^3+1} - \sqrt{x^{11}}}{\sqrt{x^2+x+1}}$

iv) Greatest Integer Function :-

Also called as Box-function or Step-up function.
Symbol $[\cdot] \equiv$ G.I.F

Ex → # $[2.72] = 2$, $[-0.73] = -1$, $[-3.74] = -4$

$\left[\frac{11}{6}\right] = 1$, $[7] = 7$



Properties of G.I.F :-

1) $[[x]] = [x]$

2) $[\text{any number}] = \text{integer}$

3) $[x \pm \text{Integer}] = [x] \pm \text{Integer}$

4) $x - 1 < [x] \leq x$

5) $[x] + [-x] = 0 \;,\; x \in \text{Integer}$
$= -1 \;,\; x \notin \text{Integer}$

6) i) $\left[\frac{x}{n}\right] + \left[\frac{x+1}{n}\right] + \left[\frac{x+2}{n}\right] + \dots + \left[\frac{x+n-1}{n}\right] = [x]$

ii) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \dots + \left[x + \frac{n-1}{n}\right] = [nx]$

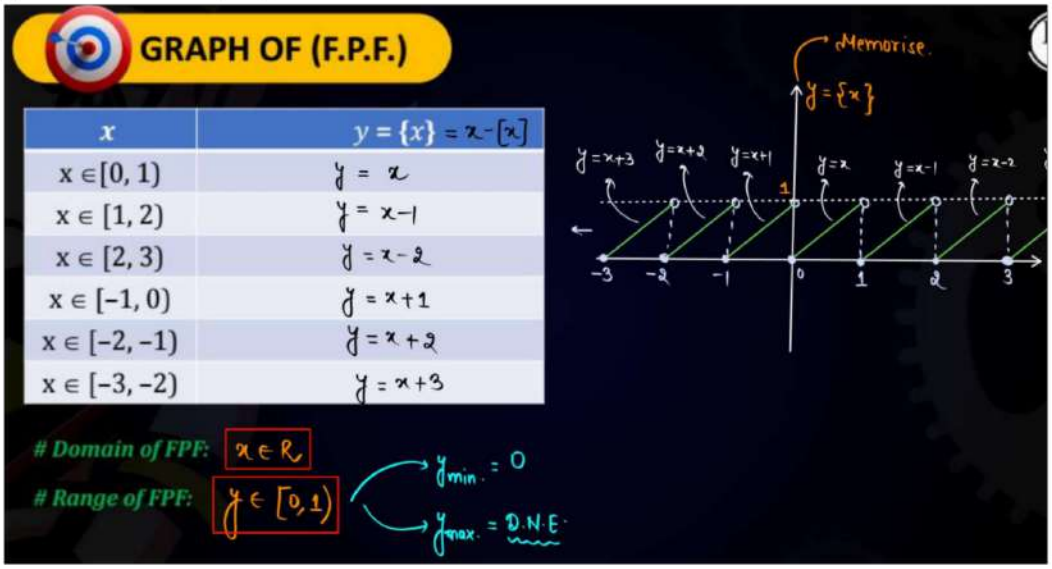
V) Fractional Part Function (FPF)

Definition : $\{x\} = x - [x]$

Ex- $\left\{\frac{11}{2}\right\} = \frac{11}{2} - \left[\frac{11}{2}\right] = \frac{11}{2} - 5 = \frac{1}{2}$

$\{\pi\} = \pi - [\pi] = \pi - 3$

$\{-3\} = -3 - [-3] = -3 + 3 = 0$



x	$y = \{x\} = x - [x]$
$x \in [0, 1)$	$y = x$
$x \in [1, 2)$	$y = x - 1$
$x \in [2, 3)$	$y = x - 2$
$x \in [-1, 0)$	$y = x + 1$
$x \in [-2, -1)$	$y = x + 2$
$x \in [-3, -2)$	$y = x + 3$

Properties of F.P.F :-

1) $0 \leq \{x\} < 1$ (Input chahe jaha, output humesha 0 & 1 ke beech hoga)

2) $\{\text{Integer}\} = 0$

3) $\{[x]\} = 0$

4) $\{x \pm \text{Integer}\} = \{x\}$

5) $\{x\} + \{-x\} = 0 \quad , x \in \text{Integer}$
$= 1 \quad , x \notin \text{Integer}$

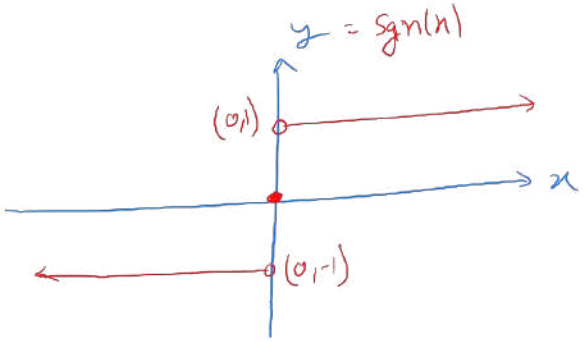
6) Solution: $\{x\} = a$; where $a \in [0,1)$
$\hookrightarrow x = \alpha + n, \; n \in \text{Integer}$

VII) Signum Function :-

$\text{Sgn}(x)$ = Signum of x

$$f(x) = \text{Sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

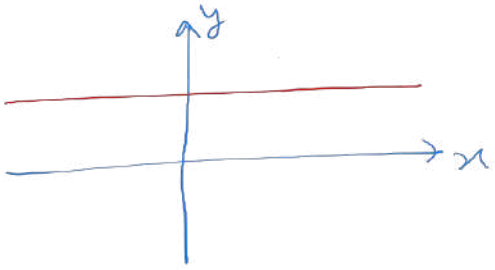
Graph :-



VIII) Constant Function :-

Output (y or f(x)) is always same.

$f(x) = K$

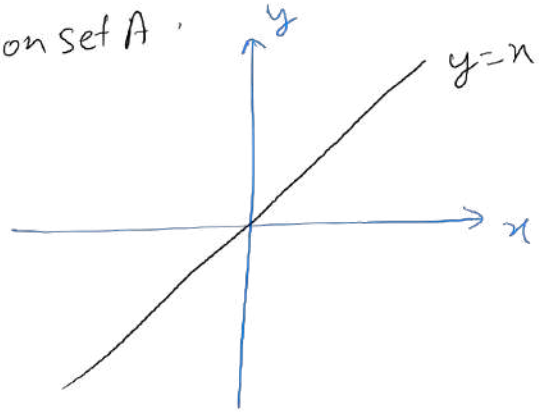


VIII) Identity Function :-

(Jo input hogi wahi output hogi)

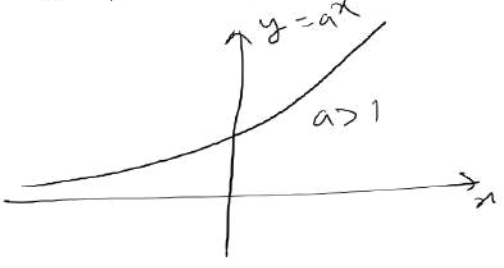
I_A = Identity function defined on set A.

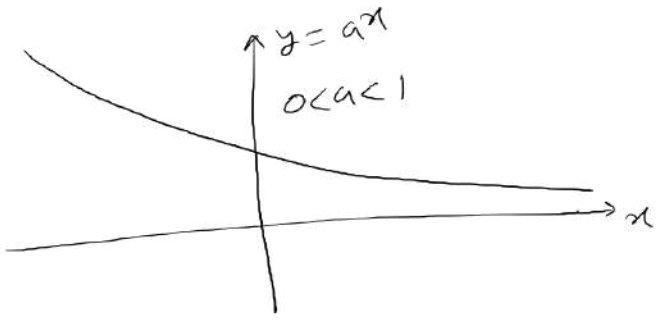
$f(x) = x$



IX) Exponential Function :-

$f(x) = a^x \; , \; a > 0, (a \neq 1)$





Most important Property :-

If $f(x+y) = f(x) \cdot f(y)$ then $f(x) = a^x$

Range :-

REMEMBER THIS TABLE

	FUNCTIONS	DOMAIN	RANGE
1.	$f(x) = a^x$	$x \in R$	$y \in (0, \infty)$
2.	$f(x) = \log_a x$	$x \in (0, \infty)$	$y \in R$
3.	$f(x) = \{x\}$	$x \in R$	$y \in [0, 1)$
4.	$f(x) = [x]$	$x \in R$	$y \in I$
5.	$f(x) = \lvert x \rvert$	$x \in R$	$y \in [0, \infty)$
6.	$f(x) = \text{sgn}(x)$	$x \in R$	$y \in \{-1, 0, 1\}$
7.	$f(x) =$ odd degree polynomial	$x \in R$	$y \in R$
8.	$f(x) = x + \frac{1}{x}$	$x \in R - \{0\}$	$y \in (-\infty, -2] \cup [2, \infty)$
9.**	$f(x) = \sin x$ } Trigonometric funct.	$x \in R$	$y \in [-1, 1]$
10.**	$f(x) = \cos x$	$x \in R$	$y \in [-1, 1]$

Methods of Finding Range :-

1) Elementary Methods of Finding Range :-

will be applicable in function (calculus) also.

Agar koi interval given ho to, uss interval mein :

1) Addition, Subtraction, Multiplication & Division by constant.
2) Reciprocal of an interval.
3) Squaring of an interval.

Squaring :

Example

Find y^2 in each case:

(i) $y \in [2, 5] \Rightarrow y^2 \in$
(ii) $y \in [-7, -3] \Rightarrow y^2 \in$
(iii) $y \in (-3, 4] \Rightarrow y^2 \in$
(iv) $y \in (-3, 2] \Rightarrow y^2 \in$
(v) $y \in [-5, 0) \Rightarrow y^2 \in$
(vi) $y \in [-7, -3] \Rightarrow y^2 \in [9, 49]$

(v) $y \in [-5, 0) \Rightarrow y^2 \in (0, 25]$

(i) $y \in \overbrace{[2, 5]}^{+ve} \Rightarrow y^2 \in [4, 25]$ ✓

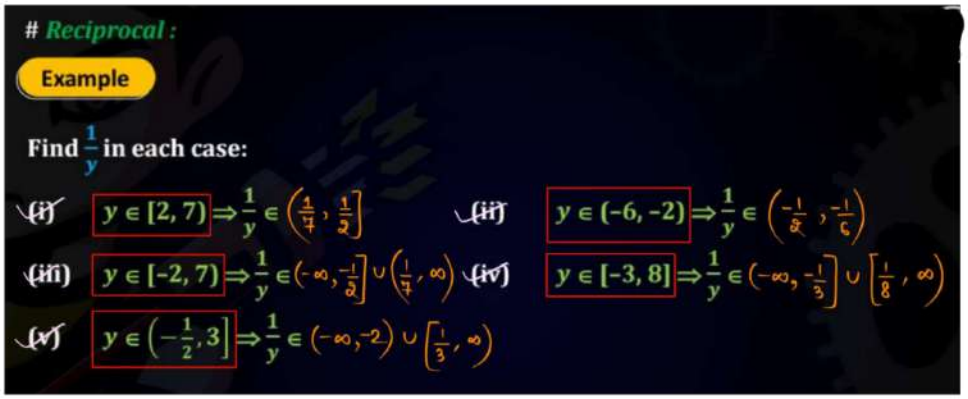
(ii) $y \in \overbrace{(-7, -3]}^{-ve} \Rightarrow y^2 \in [9, 49)$

(iii) $y \in (-3, 4] \Rightarrow y^2 \in (9, 16]$ (Wrong)

zero lies b/w interval. $\Rightarrow y^2 \in [0, 16]$ ✓

(iv) $y \in (-3, 2] \Rightarrow y^2 \in [0, 9)$

'0' ✓



2) By completing Perfect Square :-

Quadratic in terms of any function.
Perfect Square banao, elementary method again

Ex → $f(x) = x^2 + x + 3$, find range of $f(x)$.

$f(x) = x^2 + 2\left(\frac{1}{2}\right)x + \frac{1}{4} - \frac{1}{4} + 3$

$f(x) = \left(x + \frac{1}{2}\right)^2 + \frac{11}{4}$

E.M ⇒ $x \in (-\infty, \infty)$

$x + \frac{1}{2} \in (-\infty, \infty)$

$\left(x + \frac{1}{2}\right)^2 \in [0, \infty)$

$\left(x + \frac{1}{2}\right)^2 + \frac{11}{4} \in \left[\frac{11}{4}, \infty\right)$ ↓ Range

III) Converting $y = f(x)$ into $x = g(x)$:-

Ex → Find domain and Range of $f(x) = \frac{3x-2}{4-5x}$ ⟶ $4 - 5x \neq 0$

$x \neq \frac{4}{5}$

$x \in R - \{\frac{4}{5}\}$

$y = \frac{3x-2}{4-5x}$

$4y - 5xy = 3x - 2$

$4y + 2 = 3x + 5xy$

$x = \frac{4y+2}{3+5y}$

$3 + 5y \neq 0$

$y \neq -\frac{3}{5}$ } → Range $y \in R - \{-\frac{3}{5}\}$

IV) Range of composite Function :-

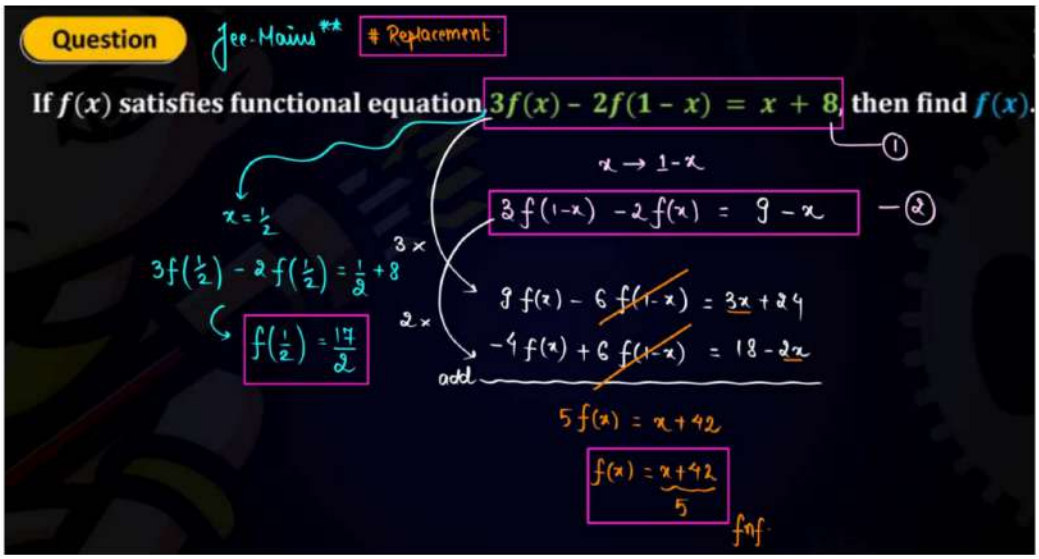
Function inside function : $y = f(g(x))$

S.I :- Calculate range of inside function $(g(x))$

S.II :- Draw graph of outside function $(f(x))$

S.III :- plot range of $g(x)$ on X-axis of $f(x)$

S.IV :- Consider graph only for shaded portion of X-axis



Solving Functional Equations :-

I) $f(xy) = f(x) + f(y)$
$f(\frac{x}{y}) = f(x) - f(y)$
$\Rightarrow f(x) = \log_a x$

II) $f(x+y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx}$

III) $f(x+y) = f(x) + f(y) \Rightarrow f(x) = kx + b$

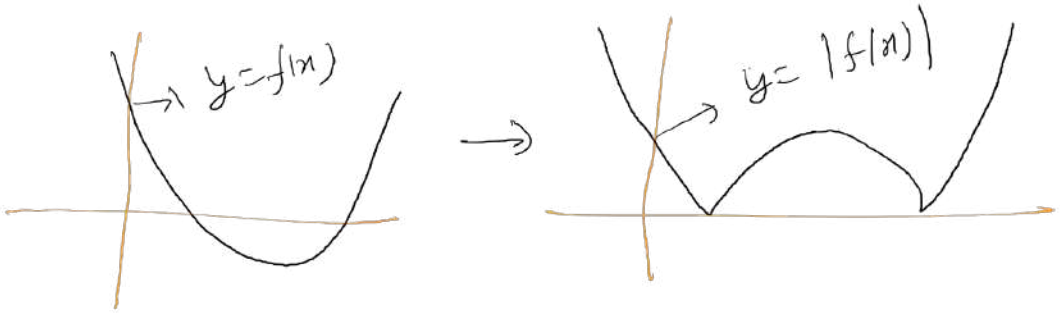
IV) $f(x)$ is a polynomial function satisfying
$f(x) f(\frac{1}{x}) = f(x) + f(\frac{1}{x}) \Rightarrow f(x) = 1 \pm x^n$

Graphical Transformation :-

If graph of $y = f(x)$ is known then :-
we can draw graph of $\Rightarrow$ $|f(x)|$, $f(|x|)$, $-f(x)$, $f(-x)$, $f(x) \pm k$, $f(x \pm k)$
$kf(x)$, $f(kx)$, $|y| = f(x)$

T-1 to draw $y = |f(x)|$
Take mirror image of graph below x-axis in x-axis and keep graph above x-axis as it is.

Ex-



T-II To draw $y = f(|x|)$:-

Remove LHS part of Y-axis, then take mirror-image RHS part in Y-axis & keep both.

T-III To draw $y = -f(x)$:-

Flip whole graph about x-axis "Niche wala upar aur upar wala niche"

T-IV To draw $y = f(-x)$:-

Flip whole graph about y-axis "Right wala left aur left wala Right"

T-V to draw $y = f(x) \pm K$:-

for $f(x)+K$:- shift whole graph by K units in upward direction (along Y-axis)
for $f(x)-K$:- shift whole graph by K unit in downward direction (along Y-axis)

T-VI To draw $y = f(x \pm K)$:-

for $f(x+K)$: shift whole graph by K units in left direction (along x-axis)
for $f(x-K)$: shift whole graph by K unit in right direction (along x-axis)

T-VII To draw $y = Kf(x)$, $K > 0$

Multiply every number on Y-axis by 'K'.
As a result :- If $K > 1$, graph will stretch along Y-axis.
If $0 < K < 1$, graph will compress along Y-axis.

T-VIII To draw $y = f(Kx)$, $K > 0$

Divide every number on x-axis by 'K'.
as a Result :- if $K > 1$, graph will compress along x-axis.
If $0 < K < 1$, graph will stretch along x-axis

T-IX To draw $|y| = f(x)$

Remove part of the graph below x-axis, now take image of portion above in x-axis & keep both.